

Intersecting Shocks: Automation, Immigration, and Their Combined Impact on Non-College-Educated Workers

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Abstract

We estimate the interaction between two key labor market forces: automation and immigration. Both shocks amplify each other, and workers exposed to automation and immigration experience significantly greater earnings losses relative to those exposed to only one. We show that while the direct effects of automation are primarily concentrated in the manufacturing industries, these area-level shocks spillover to non-college educated workers in other industries. Shifts to lower paying employers, and to some extent a shift to service occupations, are key in explaining the earnings losses among workers affected by both shocks. The interaction between shocks is quantitatively important: absent one shock, the effects of the other shock become smaller in magnitude. Our findings speak to the importance of the need for an integrated approach to understand the labor market dynamics of different labor market shocks.

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1 Introduction

Non-college educated workers are increasingly disadvantaged in the labor market compared to their college-educated counterparts. This widening disparity can be attributed to a confluence of labor market shocks that have fundamentally changed the nature of work and disproportionately impacted those without a college degree. Such shocks include the rapid acceleration of automation (Autor et al., 2003; Graetz and Michaels, 2018; Acemoglu and Restrepo, 2020; Dauth et al., 2021), the globalization of labor markets (Autor et al., 2013, 2015; Charles et al., 2018) and changes in immigration patterns (Bratsberg and Raaum, 2012; Dustmann et al., 2017). While the impacts of these factors are not necessarily unique to the non-college educated, they have overall contributed to worsening labor market prospects and wage stagnation for non-college educated workers.

While most of the previous literature analyzes these shocks in isolation, this paper underscores the importance of the interplay between various labor market shocks. Our objective is not to delineate the exhaustive interactions among all possible shocks affecting non-college educated workers. Instead, we highlight the need for an integrated approach to understanding labor market dynamics by focusing on two contemporaneous shocks—automation and immigration—for which we possess robust identification strategies. By doing so, we reveal overlooked interdependencies that can significantly influence labor market outcomes for non-college educated workers. Furthermore, our analysis contributes to the broader discourse on labor economics by illustrating how seemingly unrelated shocks can interact through spillover effects across sectors and industries. Our analysis reveals that these compounded effects can exacerbate the labor market challenges faced by non-college educated.

Using detailed administrative data from Norway, we show that automation and immigration amplify each other, and workers exposed to both shocks experience significantly greater earnings losses as a result. To identify the causal impacts of both shocks, we isolate an exogenous industry-level shock due to a sudden inflow of immigrants into specific industries following the expansion of the European Union (as in Bratsberg and Raaum, 2012) and an exogenous area-level shock due to increases in automation in other European countries (as in Acemoglu and Restrepo, 2020). Specifically, we exploit the fact that two workers who are affected by the same automation shock are differentially affected by the immigration shock, due to the inability of immigrants to work in specific jobs which had some licensing requirement. We assess the importance of the joint impact automation and immigration on labor market outcomes, comparing workers affected by both shocks to workers who are only affected by automation. Both shocks coincide with each other: industrial robots are increasing at the same time that immigration into Norway increases. The extent to which any two shocks interact with each other depends not just on how the timing of both shocks coincide with each other but also on the overlap between both shocks and the potential for spillovers between different sectors.

We show that the presence of both shocks, and providing an understanding of how shocks can amplify

each other via spillovers in the labor market, is quantitatively important. Workers can be affected by multiple shocks at the same time and, when they are, the interaction between shocks is sufficiently large to alter the overall interpretation of the estimated results. Outside of manufacturing, the negative effects of automation are driven by workers who are also exposed to the immigration shock. Among workers who are insulated from the immigration shock, there are no significant declines in earnings resulting from automation. Likewise, the negative effects of immigration originate from areas which are also highly exposed to automation. As such, our results speak to how extensive literatures on immigration and automation, which have largely overlooked the importance of the interaction between two sets of shocks, and policy should address these key forces.¹

We present three main set of results highlighting the importance of the joint impacts of automation and immigration. First, we show that while the automation shock is concentrated among the manufacturing sector, it has negative effects which spillover to all non-college educated workers in affected areas. Automation affects the primary outside option of non-college educated workers in non-manufacturing industries, as increases in industrial robots at the area-level significantly decrease earnings among manufacturing workers. Other blue-collar workers in construction experience similar declines, where earnings losses are concentrated among workers performing tasks which are directly transferable to manufacturing. Although construction workers are not directly affected by increases in industrial robots, these area-level shocks to manufacturing create spillovers which affect similar workers in other industries in the same area.

Second, we establish that the shocks of automation and immigration amplify each other, and that this interaction between shocks is sufficiently large to affect the interpretation of both shocks. Our empirical strategy compares workers similarly exposed to the automation shock, as in (Acemoglu and Restrepo, 2020), who are differentially exposed to the immigration shock, due to licensing requirements at the industry-level as in (Bratsberg and Raaum, 2012). Workers who are exposed to both automation and immigration shocks have earnings losses 2.5% greater relative to those exposed solely to automation. The joint effects of these shocks are quantitatively important in interpreting the labor market consequences of *both* shocks. Absent the immigration shock, the negative effects of automation are smaller, as workers who are insulated from immigration see no significant declines in earnings as a result of automation. Absent the automation shock, the negative effects of immigration are smaller, as earnings declines from workers exposed to immigration are concentrated in areas which are highly exposed to automation. While earnings decline among workers exposed to both shocks, employment levels remain unchanged, suggesting that the earnings decline is at least partly due to a reduction in wages. Results are robust to a host of specification checks and, importantly, we show that there exist no significant differences in the earnings between workers differentially exposed to the two shocks prior to the onset of both shocks.

¹An exception is Mandelman and Zlate (2022), who develop a theoretical model of automation, offshoring, and immigration.

Finally, we highlight the mechanisms behind why workers affected by both shocks have significantly lower earnings. Workers exposed to both shocks get stuck at the bottom, and shift to lower paying occupations in services and lower paying employers. Increases in service occupations come primarily at the expense of a decline in blue-collar occupations, which are higher paid relative to services. Workers affected by both shocks also see considerable declines in the ranking of their employer in the overall pay distribution, where one additional robot per 1000 workers among those exposed to immigration lead to a 1.3 percentile decline in their employer rank. The decline in employer rank is quantitatively important, as the move down the pay distribution corresponds to roughly a 2.5% decline in earnings. While employers at different points in the pay scale differ in a number of ways, shifting downward to a lower paid employer is sufficiently large to explain a large portion of the overall decline in earnings among workers exposed to both shocks.

Our results contribute to an extensive literature on robots and technological change (Autor et al., 2003, 2006; Goos et al., 2009; Graetz and Michaels, 2018; Acemoglu and Restrepo, 2020; Dauth et al., 2021; Acemoglu and Restrepo, 2022) as well as immigration (Borjas, 2003; Card, 2005; Ottaviano and Peri, 2012; Dustmann et al., 2016, 2017). Our findings speak to the mixed results of both literatures: Dustmann et al. (2016) reconcile some of the divergence with harmonizing the choice of empirical specification, while Jurkat et al. (2023) provide an overview of the literature on automation. We demonstrate that accounting for both shocks in tandem is quantitatively important—in a world where one shock is absent, our results show that this affects the estimated effects and overall interpretation of the other shock. As such, we highlight the need for an integrated approach in order to understand the labor market effects of both shocks. In addition, we provide an understanding of how occupational shifts and employer shifts matter for observed declines in earnings as a result of both shocks. Indeed, the existing literature emphasizes the role of both occupations (Edin et al., 2023) and employers (Abowd et al., 1999; Card et al., 2018) in determining success in the labor market. While occupations matter for effects of automation (Acemoglu and Restrepo, 2020), and they can shift as a result of both automation (Dauth et al., 2021) and immigration (Peri and Sparber, 2009), we show that the downward shift in employers is sufficiently large to explain a majority of the observed decline in earnings.

In addition, we contribute to an under-researched literature seeking to understand the dynamics of multiple shocks to the labor market. Autor et al. (2015) emphasize the importance of trade and technology, finding that Chinese import competition and computerization both impact labor markets, but in distinct ways. In contrast, we demonstrate how the joint shocks of immigration and automation can interact with each other, which affects the interpretation of both shocks. Goos et al. (2014) develop a theoretical model to understand jointly the impacts of routine biased technical change and offshoring, while Mandelman and Zlate (2022) develop a structural model to understand jointly offshoring, automation, and immigration. Our results exploit exogenous variation in both automation and immigration

shocks to understand the labor market effects over time.

2 Data

We make use of detailed administrative data from Norway, which contains data on demographic characteristics, employment, earnings, education. We measure a worker’s exposure to industrial robots using data from the International Federation of Robotics (IFR) at the local labor market level, and a worker’s exposure to immigration at the industry-level measured before the immigration shock.

2.1 Norwegian register data

First, the population register provides data on demographic characteristics such as birth year, municipality of residence, country of birth, and gender. We map municipalities to local labor markets as in Gundersen and Juvkam (2013), and use country of birth to distinguish between native and immigrant workers.

Second, data from the employer register provides detailed annual information on employment status, industry of employment, and the plant and firm at which a worker is employed. We also observe a worker’s occupation from 2003, and classify occupations into four categories: professional, blue-collar, services, and elementary occupations. Service occupations are dominated by shop salespersons and protective services, while elementary occupations comprise office cleaners and supporting workers in maintenance and construction. Blue-collar occupations are dominated by craft, trade, and machinery occupations, while professional occupations comprise managerial work, professionals, and technicians. Data on earnings is also available, and throughout the paper, we measure a worker’s gross annual labor market earnings, which includes any business income earned throughout the year.

Finally, data on an individual’s educational attainment is extracted from the education register. We observe any ongoing study as well as any qualification attained. We define college educated workers as those with any post-secondary education, including degrees from universities, university colleges, and technical colleges. As such, non-college educated workers include both high school graduates and high school dropouts.

2.2 IFR Data

Data on industrial robots is measured by the IFR from 1993–2015. Our measure of industrial robots closely follows that of Acemoglu and Restrepo (2020), and we match the industry in which robots are installed at the broad industry level, roughly 2 digit NACE with 19 industries. Throughout the paper, we measure the average yearly change at the local labor market level in industrial robots in five European countries—Denmark, Finland, France, Italy, and Sweden—as our measure of an industry’s exposure to

industrial robots, as defined below in Section 3.

3 Empirical Framework

Understanding the causal impact of automation and immigration on labor market outcomes requires isolating exogenous variations in these shocks. In this section, we outline our identification strategy for capturing the exogenous exposure to both automation and immigration, leveraging natural experiments and methodologies established in the literature.

3.1 Automation Shock

To isolate the impact of automation on labor market outcomes, we follow Acemoglu and Restrepo (2020) and employ a shift-share approach to estimate the predicted penetration of robots in a given local labor market. This measure predicts how much robot adoption in other countries would affect a local labor market based on its initial industry composition. Local labor markets with a higher concentration of industries experiencing rapid automation internationally are predicted to have higher exposure to automation. Because we rely on international (rather than Norwegian) trends in robot adoption, we mitigating endogeneity concerns and isolate exogenous variation in automation that is not driven by local economic conditions.

First, we calculate the adjusted penetration of robots at the industry level using data from five other European countries (Denmark, Finland, France, Italy, and Sweden). By averaging the change in robot usage per worker across multiple countries, we obtain an industry-level measure of automation that reflects international trends rather than domestic factors. The formula for the adjusted robot penetration in industry j is given by:

$$\overline{\text{adjusted penetration robots}_j} = \frac{1}{5} \sum_c \frac{M_{j,e}^c - M_{j,s}^c}{L_{j,s}^c} \quad (1)$$

where: $M_{j,e}^c$ is the number of robots in industry j in country c at the end of the period, $M_{j,s}^c$ is the number of robots in industry j in country c at the start of the period, and $L_{j,s}^c$ is the baseline employment in industry j in country c at the start of the period. Throughout the paper, we measure the change in robots from 1996–2015.

We then use these national industry-level shifts in predicted robot adoption to compute predicted robot adoption in each local labor market, using the shift-share methodology. This approach makes use of the fact that local labor markets differ in their industrial composition (as measured by initial employment shares), but are assumed to be similarly affected by national or international industry-level shocks. We calculate predicted robot penetration in local labor market m as:

$$\Delta \text{Automation}_m = \sum_j \left(\frac{L_{j,m,2000}}{L_{m,2000}} \times \overline{\text{adjusted robot penetration}_j} \right), \quad (2)$$

where: $L_{j,m,0}$ is the baseline employment in industry j in local labor market m at the start of the period, $L_{m,0}$ is the total baseline employment in local labor market m at the start of the period, and $\overline{\text{adjusted robot penetration}_j}$ is the adjusted robot penetration for industry j from Equation 1.

3.2 Immigration Shock

To isolate the exogenous variation in immigration, we exploit the expansion of the European Union (EU) in 2004, which significantly altered migration flows into Norway. Although Norway is not an EU member state, it is part of the European Economic Area (EEA), which allows for the free movement of goods, services, capital, and people between EU and EEA countries. The 2004 enlargement of the EU included eight Central and Eastern European countries: the Czech Republic, Estonia, Hungary, Latvia, Lithuania, Poland, Slovakia, and Slovenia. Prior to 2004, immigration from these countries to Norway was relatively restricted due to stringent immigration policies. The EU expansion led to the establishment of a common European labor market, enabling workers from the new member states to access the Norwegian labor market more easily (Bratsberg and Raaum, 2012). This policy change resulted in a sudden influx of immigrant workers.

An important institutional feature of the Norwegian setting is that certain industries have strict licensing requirements that pose barriers to entry for foreign workers. For example, some construction jobs require specific certifications or licenses that are contingent on completing Norwegian qualifications, and as such are time-consuming and costly to obtain for new immigrants. Consequently, immigrant workers disproportionately entered unlicensed sectors within the construction industry, such as general labor or non-specialized trades, while licensed sectors remained relatively insulated from this influx (Bratsberg and Raaum, 2012).

When investigating the impact of exposure to immigration, we define our treated group as workers who were employed in unlicensed construction industries in 2000, prior to the EU expansion. The control group consists of similar workers employed in licensed construction industries. This distinction allows us to compare labor market outcomes between workers directly affected by the immigration shock and those who were not, within the same geographic and economic environment.

3.3 Regression Models

To investigate the impact of these automation and immigration shocks on individual labor market outcomes, we employ regression models that relate changes in individual outcomes to shock exposure. We start by identifying the effect of exposure to one shock—automation—in isolation. Specifically, we esti-

mate the following baseline model:

$$\Delta y_i = \beta_0 + \beta_1 \Delta Auto_{m(i)} + \gamma_{c(i)} + \lambda_{e(i)} + \varepsilon_i \quad (3)$$

where: Δy_i is the change in a given outcome for individual i from 2000 to 2015 and $\Delta Auto_{m(i)}$ is the predicted change in automation exposure in the local labor market m where individual i resides, as measured in equation (1). $\gamma_{c(i)}$ represents cohort fixed effects for the birth cohort c of individual i , while $\lambda_{e(i)}$ denotes education fixed effects for the education level e of individual i . ε_i is the error term. Throughout, standard errors are clustered at the initial local labor market level. Equation 3 captures the average effect of automation exposure on changes in outcomes across all individuals, controlling for cohort and education effects.

We then contrasts these effects of exposure to automation with estimates of joint exposure to both automation and the intersecting shock of immigration. To estimate the differential impact of automation on workers affected by the immigration shock, we extend the model by introducing an interaction term between automation exposure and an indicator variable for being treated by the immigration shock. This model compares unlicensed and licensed construction workers within the same area. Specifically, we estimate the following model:

$$\Delta y_i = \delta_0 + \delta_1 \Delta Auto_{m(i)} + \delta_2 ImmExp_i + \delta_3 \Delta Auto_{m(i)} \times ImmExp_i + \gamma_{c(i)} + \lambda_{e(i)} + \varepsilon_i \quad (4)$$

where: $ImmExp_i$ is an indicator variable equal to 1 if individual i was employed in unlicensed construction industries in 2000 (thus directly affected by the immigration shock), and 0 if employed in licensed construction industries. Unlicensed workers are more likely to have been affected by the influx of immigrants following the 2004 EU expansion, as unlicensed industries faced fewer barriers to entry for foreign workers. In contrast, licensed construction industries required specific certifications, limiting the impact of immigration on those sectors. In our final sample, there is an approximately equal split between workers in unlicensed and licensed industries.

In Equation 4, δ_1 measures the effect of automation exposure on earnings changes for licensed construction workers who are not exposed to immigration. δ_2 captures earnings changes between those exposed vs. not exposed to immigration due to licensing protection. Both δ_1 and δ_2 are interpreted as the effect of one shock, in the absence of the other shock. For instance, δ_1 estimates the change in earnings among workers who are affected by automation, but are insulated from the immigration shock. Similarly, δ_2 estimates the change in earnings among those exposed to the immigration shock, absent

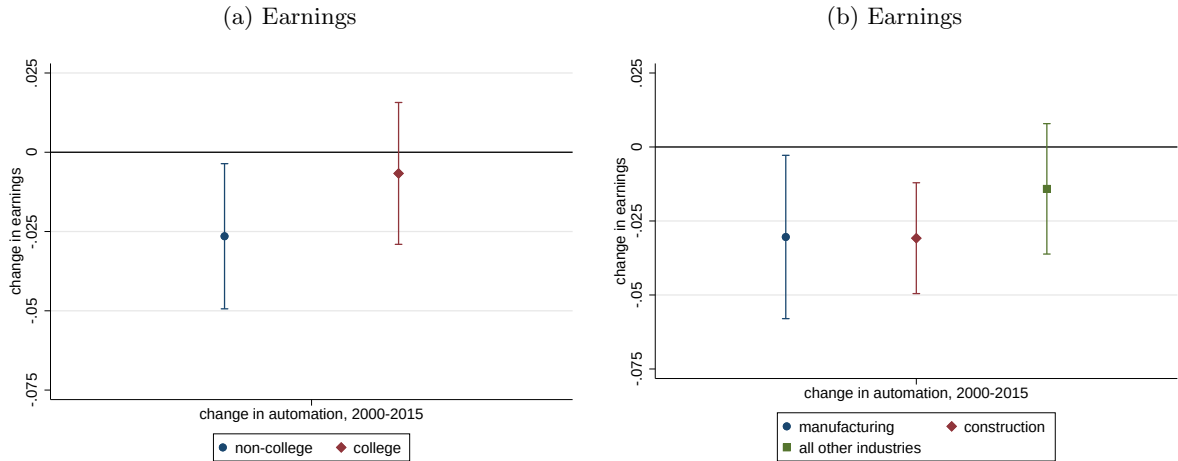
any change in the automation shock.

By including the interaction term between both shocks, we assess whether the impact of automation differs for workers who were also exposed to the immigration shock. The coefficient on the interaction term $\Delta\text{Auto}_{m(i)} \times \text{ImmExp}_i$, δ_3 , captures the additional effect of automation for those exposed to immigration compared to those not exposed to immigration due to licensing requirements. When looking at earnings as the outcome, a negative and significant δ_3 would indicate that automation has a more adverse effect on earnings for those simultaneously exposed to immigration.

4 The Impacts of Automation and the Importance of Industry Spillovers

Figure 1 illustrates estimates of equation (3), the impact of automation on earnings for the full population of male workers aged 18 to 49 in 2000.² Panel (a) clearly demonstrates that the negative consequences are concentrated among non-college-educated workers. Specifically, while there are no significant changes in the earnings of college-educated workers, one additional robot per 1000 workers reduces earnings by 2.5% among non-college educated workers. This represents a large decline in earnings: Acemoglu and Restrepo (2020) find that men with less than college education experience declines in hourly wages of 0.75–1.1% as a result of a similar change.

Figure 1: The Impacts of Automation on Earnings by Industry/Education, from 2000–2015



Notes: sample of men aged 18–49 employed initially in 2000. Figure presents estimates of equation (3), estimated separately for workers with different education levels (panel a) and workers initially employed in different industries (panel b).

Interestingly, the impact of automation on the earnings of male workers in the local labor market is not confined to the manufacturing industry, where automation predominantly occurs. As shown in Panel (b), the negative impact on workers in the manufacturing industry is mirrored by a similar impact

²As shown in Figure B.2, the earnings impact of automation is concentrated among men. There is no effect on the earnings of women.

on workers in the construction industry. One additional robot per 1000 workers lowers the earnings of those initially employed in manufacturing and construction by roughly 2.5%. Contrastingly, there is no statistically significant impact on earnings for workers in all other industries. The declines in earnings are not accompanied by declining employment rates for any workers (Figure B.3), suggesting that the overall earnings declines is at least partially due to reductions in relative wages over time.

While it is unsurprising that the automation shock affects workers in the directly impacted industry of manufacturing, the spillover effect to workers in construction warrants further investigation. Table C.1 highlights the considerable similarities between manufacturing and construction workers. These workers are comparable in terms of demographic characteristics, earnings, and education levels, with approximately 85–90% being non-college educated. In contrast, there exist considerable differences between those who work in all other industries and workers in construction or manufacturing. Less than 70% of workers in all other industries are college educated.

These spillover effects are likely driven by the fact that workers flow between different industries and that manufacturing represents an outside option for some workers in construction. To shed further light on this, we examine whether the spillovers we observe are driven by construction workers who are more likely to shift into manufacturing before the automation shock. We start by predicting the probability that a worker transitions to manufacturing from another industry based on individual characteristics, age and age squared, and four task measures: routine, service, math, and social tasks.³

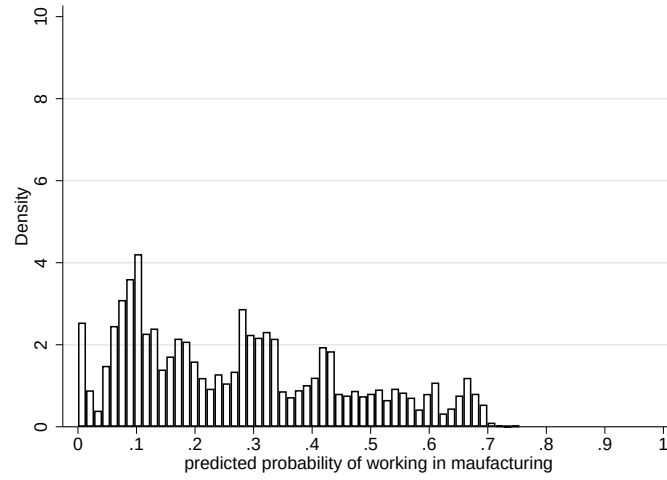
Figure 2 presents our findings. Panels (a) and (b) show the distributions of the predicted probability of shifting to manufacturing among non-construction workers and construction workers, respectively. The level and dispersion of the distribution informs the transferability of tasks among both sets of workers. The distributions indicate that construction workers, on average, have a higher predicted probability of shifting to manufacturing than non-construction workers. While the distribution for construction workers is centered around 0.5 and relatively evenly dispersed, the distribution for non-construction workers is concentrated around lower probabilities. While a considerable fraction of construction workers have predicted probabilities of shifting to manufacturing of 0.5 and above, the same cannot be said for non-construction workers. Furthermore, the distribution in Panel (b) shows that certain construction workers have a particularly high likelihood of transitioning to manufacturing based on their characteristics and task profiles.

Panel (c) of Figure 2 illustrates the differential impacts of automation on construction workers, dividing them into two groups: those above and those below the median probability of shifting to manufacturing. The results reveal that the negative earnings effect of automation is concentrated among construction workers with a higher probability of transitioning to manufacturing. These findings suggest that the spillover effects of automation are concentrated among workers employed in similar industries

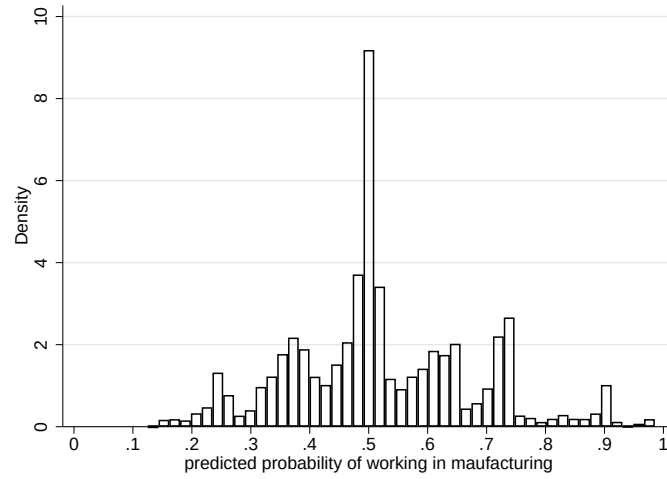
³The task measures are obtained from occupational data in 2003, the first year when such detailed information is available. We link occupations to tasks as in Deming (2017), linking occupations in Norway to task measures as in O*NET.

Figure 2: The Differential Impacts of Automation Across Different Types of Non-Manufacturing Workers

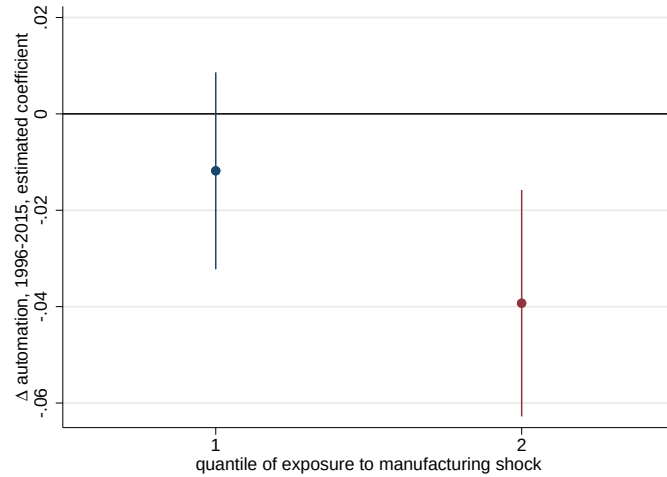
(a) Distribution of Probability of Shifting to Manufacturing, Non-Construction



(b) Distribution of Probability of Shifting to Manufacturing, Construction



(c) Differential Impacts of Automation by Probability of Shifting to Manufacturing, Construction Workers



Notes: panel (a) plots the distribution of the probability of shifting to manufacturing among all non-construction employees. Panel (b) plots the distribution of the probability of shifting to manufacturing among all construction employees. Panel (c) plots the impact of automation among construction workers, separately by whether they are above/below the median probability of shifting to manufacturing. Sample of men aged 18–49 employed initially in 2000. Tasks are measured in 2003, the first year when data on occupations is available.

who possess comparable skills and perform similar tasks. The mechanism behind this spillover may involve increased competition for jobs, as automation displaces manufacturing workers who then seek employment in construction, thereby increasing the labor supply and exerting downward pressure on wages. At the same time, this lowers the outside option available to construction workers with comparable skills to those employed in manufacturing.

Our analysis underscores the importance of considering cross-industry linkages and worker substitutability when evaluating the broader impact of automation on local labor markets. Policies aimed at mitigating the adverse effects of automation should account for these spillover effects to effectively support vulnerable worker groups.

5 The interaction between automation and immigration shocks

The preceding analysis highlights the significant impact of automation on non-college-educated male workers, not only within the manufacturing industry but also through spillover effects to the construction industry. This finding underscores the interconnectedness of industries and the importance of worker substitutability in understanding labor market dynamics. However, we argue that focusing solely on the impact of automation may overlook other concurrent factors influencing these workers' outcomes. In particular, the same period saw substantial immigration flows into Norway following the 2004 EU expansion, predominantly affecting non-college-educated workers in industries like construction (see Figure 3a). The increase in immigration is overwhelmingly comprised of non-college educated workers, who outnumber college educated immigrants nearly 4:1 (Figure 3b). Analyzing these contemporaneous shocks in isolation may lead to incomplete or misleading conclusions. In this section, we therefore explore the interaction effects between automation and immigration, demonstrating the importance of considering multiple, concurrent labor market shocks to fully understand their combined impact on non-college educated workers.

The impacts of automation on earnings are concentrated among construction and manufacturing workers. While automation increases considerably after 2000 (see Figure A.1), so too does immigration after the EU's expansion in 2004. There exists considerable overlap at the industry-level between the two shocks of automation and immigration, as two of the industries which see some of the largest increases in immigration are construction closely followed by manufacturing (Figure 3a).

Similar to automation, Figure 3 shows that the resulting increase in immigration after EU's expansion was overwhelmingly concentrated among low-educated workers in specific industries. To further understand the interaction between automation and immigration, we focus on a sample of construction workers. The inflow of immigrants into construction jobs was considerable, and immigrants from 2004 expansion countries represent around 15% of all construction workers by 2017, compared to less than

5% of non-construction industries (Figure 3c). Within all construction workers, our empirical strategy compares the differential impacts of automation among treated workers (those working in unlicensed construction industries) and counterfactual workers (those working in licensed construction industries). As described in Section 3, licensing requirements in certain construction industries acted as a barrier to employment for immigrants. Figure 3e reveals the important role of licensing requirements in restricting immigration: while immigrants comprise 7% of employment in licensed industries, they make up 22% of unlicensed construction workers by 2017. Both industries had similar immigrant shares close to zero prior to the expansion of the EU, underlying the importance of the EU’s expansion for immigration.

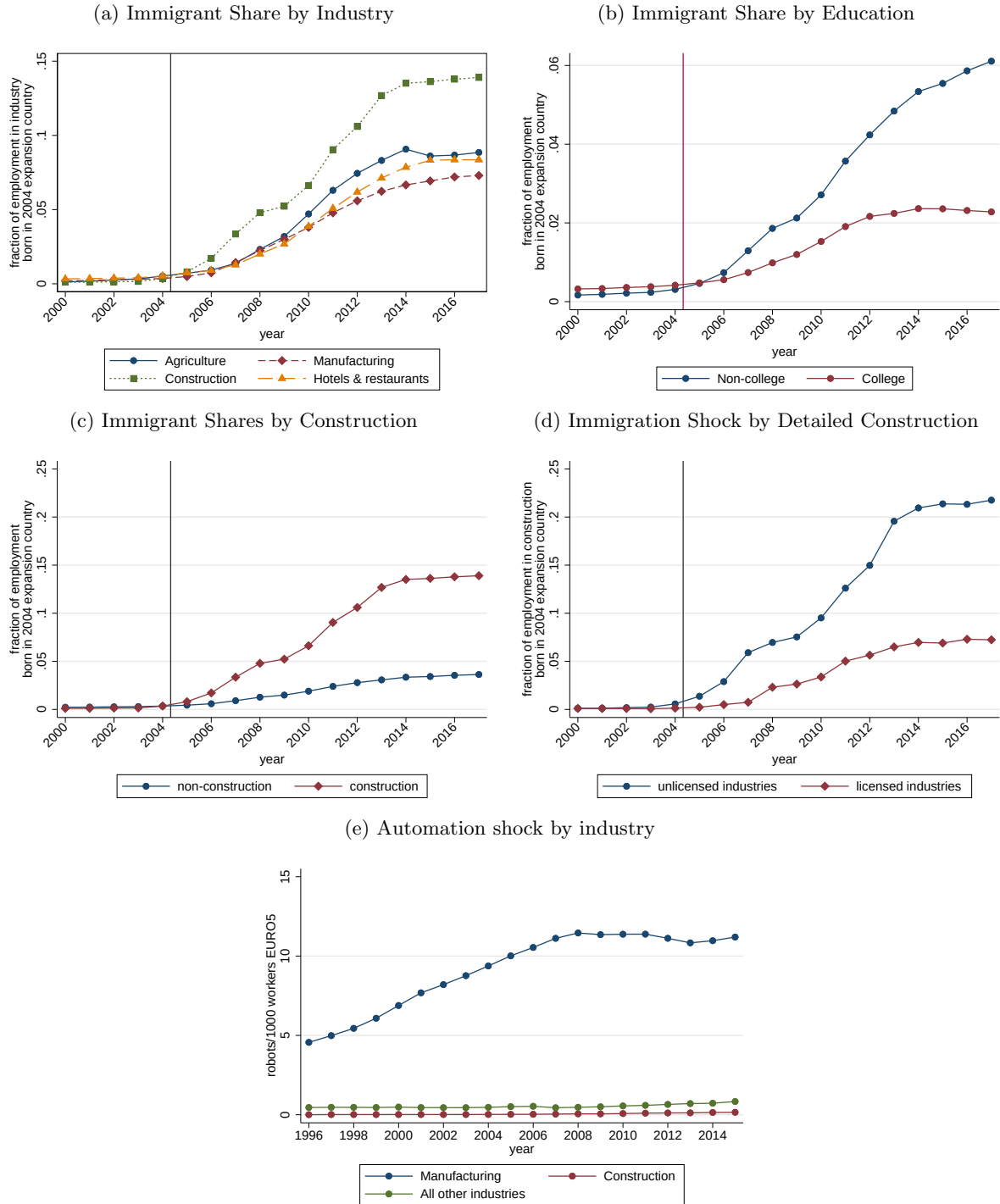
Figure D.4 shows the distribution of the automation shock from 1996–2015 across different local labor markets in Norway. While there are some pockets of areas with larger increases in industrial robots, the shock is fairly spread across different regions of Norway. While all areas see increases in industrial robots, some local labor markets see smaller increases and some see much larger increases. Our empirical strategy exploits two different sources of variation: shifts in automation at the local labor market level and shifts in immigration at the industry level. While these shocks occur at different levels, it is useful to understand the overlap between the two of them at the area-level. Figure D.5 presents the correlation between the change in immigration and automation at the local labor market level. There is no systematic relationship between the immigration shock and automation shock, and places which are differentially affected by the automation shock see similar changes in immigration. This suggests that, at the area-level, the automation shock does not predict changes in the patterns of immigration.

5.1 The Intersection Between Area-Level Automation and Industry-Level Immigration Shocks

Table 1 estimates the differential impact of automation, comparing the effects of automation among workers exposed to the immigration shock to those insulated from immigration in a triple difference specification. Focusing on interaction between treatment and the change in automation reveals that the earnings losses resulting from automation are significantly worse among construction workers who are also treated by the immigration shock (column 1). For such workers, an increase of one robot per 1000 workers decreases earnings by 2.5%, over and above the effects of automation on counterfactual construction workers.

While the combined effect of both shocks leads to even larger declines in earnings, a remaining question is how important are the effects of each shock in isolation. The non-interacted coefficient of the change in automation provides the effects of automation on workers not treated by the immigration shock. Among workers only affected by automation and not the immigration shock, there are negative but non-significant declines in earnings of 1.4%. This suggests that absent the immigration shock, the effects of automation are considerably smaller in magnitude. Likewise, the non-interacted coefficient of unlicensed

Figure 3: Describing the Immigration and Automation Shocks over time and by Industry



Notes: Panel (a) reports the share of immigrants employed in the top 4 industries affected by immigration. Panel (b) reports the share of immigrants employed in any industry, separately by college/non-college educated. Panel (c) reports the share of immigrants employed in construction and non-construction industries. Panel (d) reports, within the construction industry, the share of immigrants who are employed in either unlicensed or licensed industries. Panel (e) reports the evolution of automation as defined in equation (1), separately by manufacturing, construction, and all other industries.

Table 1: The Interaction Between Automation & Immigration

	Δ Log Earnings		
	(1) Full Sample	(2) Licensed	(3) Unlicensed
Δ automation, 1996-2015	-0.014 (0.015)	-0.014 (0.015)	-0.036*** (0.010)
Unlicensed 2000	-0.024 (0.021)		
Δ automation $\times$ unlicensed 2000	-0.024** (0.011)		
Education FE (1 digit)	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes
Individuals	82459	41533	40926
Average x	1.328		

Notes: Columns (1) reports estimates from equation (4), for earnings. Columns (2)–(3) report estimates from equation (3), separately for licensed and unlicensed workers, for earnings. Standard errors are clustered at the initial local labor market level. Standard errors are clustered at the initial local labor market level. ***, **, and * correspond to significance at the 1%, 5%, and 10% levels respectively.

workers reveals the effects of immigration on changes in earnings, absent any change in automation. As seen with the automation shock, the effects of immigration are negative, but not statistically significant, suggesting the decline in earnings among workers treated by immigration is originating from areas which are also exposed to automation.

Compared to treated workers, the estimated base coefficient for automation reveals that counterfactual construction workers see negative impacts on their earnings, but no significant change. Such stark differences among workers employed in construction suggest that the shocks of automation and immigration interact considerably with each other. This is also seen in columns (2) and (3), which estimates separate difference-in-differences regressions between treated and counterfactual construction workers. While there are no significant changes in the earnings of counterfactual workers as a result of automation, treated workers experience significant declines in earnings of nearly 4%. Consistent with previous results, declines in earnings are not driven by any significant changes in employment over time (Table E.2).

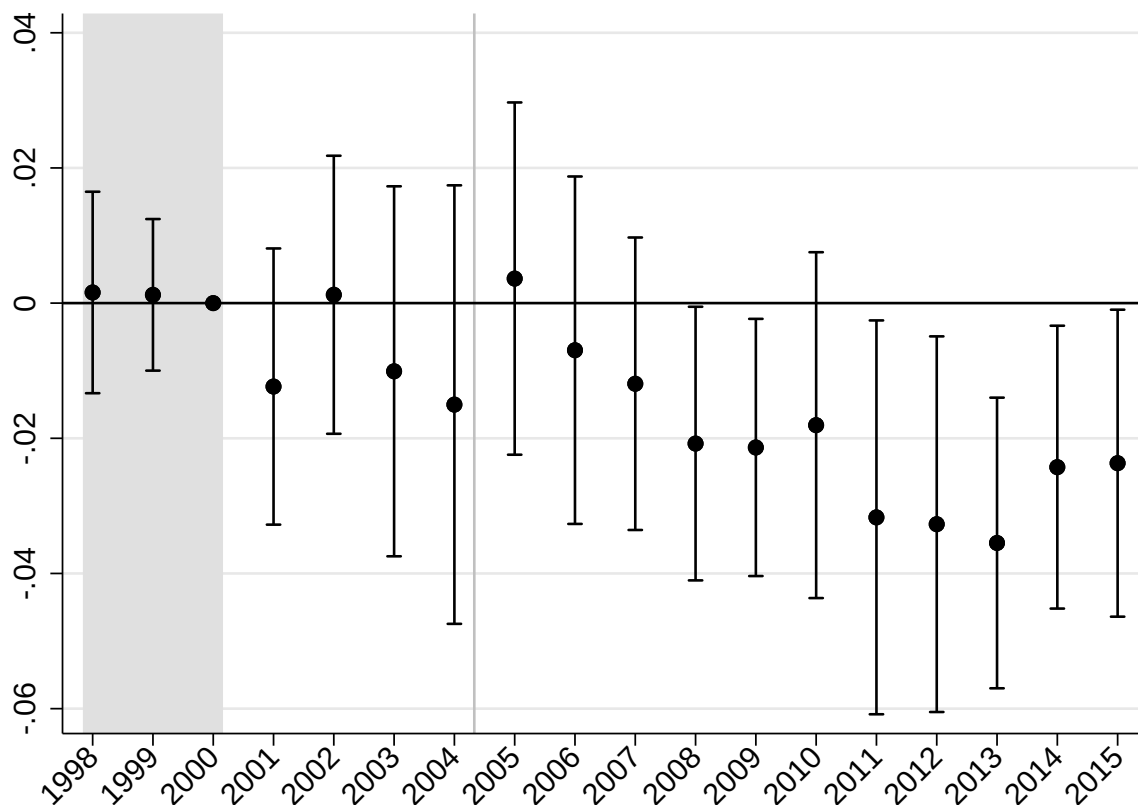
5.2 Validating the Empirical Design and Robustness

Figure 4 estimates the differential effects of automation between unlicensed and licensed workers, fixing the shock to the change in automation from 1996–2015 but allowing the estimated earnings effect to vary over time from 1998–2015.⁴ Earnings become negative and significant as automation in EURO5 countries hits its maximum and plateaus in 2008 (as seen in Figure A.1). Prior to the immigration shock in 2004, estimated coefficients are flat and not significantly different from zero. Such similarity suggests

⁴Results are similar allowing the change in automation to evolve flexibly from year to year.

that workers treated by both shocks are on a similar earnings trajectories to those treated by just one shock prior to the onset of both shocks. Figure F.7 goes even further back in time, trading off losing some of the sample with a longer pre-period. Very similar to what is seen in Figure 4, the figure suggests that unlicensed/licensed workers who are differentially impacted by automation do not have differential trends prior to the immigration shock.

Figure 4: The Interaction Between Automation & Immigration, Over Time



Notes: Figure reports estimates from equation (4), with outcome variables measured at the change in earnings for each year relative to 2000. Regressions are estimated separately year-by-year, such that the value for 2000 is not estimated. Vertical line at 2004 represents when EU expansion occurs. Change in automation is measured as the change in automation from 1996–2015 for all years, such that the value of the automation shock is fixed over time.

Reassuringly, there are not differential pre-trends in the differential impacts of automation between unlicensed and licensed workers. However, this differential exposure to automation may be driven by particular areas or the shocks which we exploit may be correlated with other underlying trends in the labor market. Table 2 reports a series robustness checks, estimating the differential impacts of automation from 1996–2015 across several different specifications.

First, column (1) shows that results are not driven by areas with very large changes in automation, and results are slightly larger when excluding local labor markets in the the top 1% of the change in automation. Second, column (2) shows that other trends in the manufacturing industry do not drive the differential exposure to automation. By controlling for the initial share of employment in

manufacturing, the regression estimated in differences allows for differential trends in earnings across areas with different initial levels of manufacturing. Third, while our results find that the shocks of immigration and automation interact, Chinese imports were also increasing considerably in Norway over the same period. Column (3) shows that controlling for Chinese imports from 1996–2007 as in Balsvik et al. (2015) has no discernible impact on our results, suggesting that the results we attribute to the intersection between automation and immigration are not confounded by a third shock.

Fourth, columns (4) and (5) control for demographic characteristics—factors such as the initial share of females, college educated workers, Norwegians, and workers over 50—and population density respectively. As with controlling for industry factors, results are robust to controlling for differential trends in demographics and population. Fifth, column (6) controls for the initial share of workers who are union members, which may be correlated with trends in earnings across areas which are differentially affected by the automation and immigration shocks. As seen before, results are unchanged allowing for such trends. Finally, column (7) includes all controls included separately in columns (2)–(6) in the same regression, and again results are similar to adding these controls progressively.

Table 2: The Interaction Between Automation & Immigration, Robustness

	Excluding High Exposure	Additional Controls					
	(1) Δ Log Earnings	(2) Δ Log Earnings	(3) Δ Log Earnings	(4) Δ Log Earnings	(5) Δ Log Earnings	(6) Δ Log Earnings	(7) Δ Log Earnings
Δ automation $\times$ unlicensed 2000	-0.0277** (0.0129)	-0.0233** (0.0114)	-0.0238** (0.0115)	-0.0226** (0.0113)	-0.0227** (0.0111)	-0.0233** (0.0114)	-0.0223** (0.0113)
Manufacturing share	No	Yes	No	No	No	No	Yes
Exposure to Chinese Imports	No	No	Yes	No	No	No	Yes
Demographic Controls	No	No	No	Yes	No	No	Yes
Union Density	No	No	No	No	No	Yes	Yes
Population Controls	No	No	No	No	Yes	No	Yes
Observations	81736	82459	82459	82459	82459	82459	82459

Notes: Table reports estimates from equation (4), where each column corresponds to a separate regression for each robustness check. Column (1) excludes the places with the highest rates of automation. Columns (2)–(6) add one control variable in each model, while column (7) includes all of the variables added from columns (2)–(6) in the same regression. Standard errors are clustered at the initial local labor market level. ***, **, and * correspond to significance at the 1%, 5%, and 10% levels respectively.

6 Understanding the response of unlicensed construction workers

Our previous results demonstrate how the effects of automation are significantly worse among workers who are also treated by an immigration shock. In this section, we examine the underlying mechanisms behind why workers affected by a tandem of shocks lose out comparatively more. Our previous results also establish that employment does not differentially change among workers affected by both the automation

and immigration shocks. As such, any response from affected workers will be conditional on employment, such as the type of job a worker is employed in or the employer at which a worker is employed.

The existing literature emphasizes the importance of occupations for both automation and immigration shocks. Acemoglu and Restrepo (2020) show that the negative effects of automation are concentrated among blue-collar occupations, which are more susceptible to the displacement effects of automation. As seen previously, this is also the case in Norway, and this dramatically decreases the outside option of workers in our sample. At the same time, Dauth et al. (2021) show considerable reallocation effects of automation, with a decline in manufacturing jobs which is met with an equal increase in reallocation of jobs to services. Indeed, occupational decline harms workers (Edin et al., 2023), and an extensive literature on skill biased technical change demonstrates the collapse of mid-skilled occupations (Autor et al., 2006; Goos et al., 2009). Similarly, immigration shocks lead to occupational shifts among natives. Peri and Sparber (2009) show natives shift to occupations which specialize in communication and language tasks as a result of increases in immigration, and as immigrants into construction are directly competing with natives (Bratsberg and Raaum, 2012), natives may seek employment opportunities which specialize in other tasks where they have a comparative advantage relative to immigrants.

We find that both shifts in occupations and employers are important in explaining earnings losses. Workers shift occupations down the job ladder into lower paid service occupations. At the same time, workers also shift to lower wage employers, moving down the pay scale to plants which are lower ranked in the distribution.

6.1 Worker-level Response: Occupational Shifts

Table 3 presents the additional change in occupations among workers who are affected by both the automation and immigration shock. Occupations are only available from 2003, and the table presents changes in occupations from 2003–2015.⁵ Occupations are classified into four broad groups: professional (usually white-collar) occupations, blue collar occupations, service occupations, and elementary occupations. Service occupations are dominated by shop salespersons and protective services, while elementary occupations comprise office cleaners and supporting workers in maintenance and construction.

Workers affected by both shocks step down the job ladder, and see significant increases in the probability of working in service occupations (column 1). Increases in services come primarily at the expense of blue-collar occupations (column 3) and to some extent professional occupations (column 2). The shift to service occupations is considerable, where one additional robot per 1000 workers increases the probability of working in services by 1 percentage point. Among our sample, only 5.5% of workers are employed in services by 2003, corresponding to an increase of 18%.

Service occupations are lower paid relative to both blue-collar: among all workers, men employed in

⁵Table 3 restricts the sample to those who have an occupation defined in 2003, as not all workers who are employed have an occupation reported at the start of the sample.

service occupations earn roughly 35% less. In a back of the envelope calculation, we can quantify how much of an earnings decline is predicted by shifting from blue-collar occupations to services. Taking this gap in earnings between services and blue-collar work, a 1 percentage point increase in services from blue-collar occupations would explain around 15% of the total decline in earnings in column (1) of Table 1 ($\frac{0.01*-0.35}{-0.024} = 0.146$). As such, the shift down the job ladder represents one important reason behind why both shocks make workers worse off.

Table 3: The Interaction Between Automation & Immigration, Occupational Shifts

	(1) Δ Service	(2) Δ Professional	(3) Δ Blue Collar	(4) Δ Elementary
Δ automation $\times$ unlicensed 2000	0.010*** (0.004)	-0.006 (0.005)	-0.014 (0.009)	0.003 (0.003)
Education FE (1 digit)	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes
Individuals	70047	70047	70047	70047
Average x	1.326	1.326	1.326	1.326

Notes: Table reports estimates from equation (4), with the outcome variable as the change in occupation. Changes in occupations are measured from 2003–2015, where 2003 is the first year which occupation data is available. Standard errors are clustered at the initial local labor market level. ***, **, and * correspond to significance at the 1%, 5%, and 10% levels respectively.

6.2 Worker-level Response: Shifting Employers

Workers affected by both shocks reallocate their labor supply to specialize in occupations performing service tasks. While such occupations tend to be lower paid, and can explain a portion of the observed decline in earnings, the majority of this decline is unexplained by occupational shifts. We examine the importance of changes in plant-level factors, calculating the percentile rank of each plant in Norway using the median earnings at the plant-level.

Table 4 shows that workers affected by both shocks also shift their employer, and experience a step down the ranking of their employer. As a result of both shocks, affected workers move to a plant which is 1.3 percentiles lower (column 1). This decrease is similar when calculating percentile ranks using just native workers (column 2), suggesting that this decline in employer rank is not the direct result of immigrants working in the same employer. The downward shift in employer pay is quantitatively important: moving one percentile rank down the pay distribution corresponds to roughly a 2% decline in median earnings. While employers at different points in the pay scale differ in a number of ways, the downward shift to lower wage employers is sufficiently large to explain a large portion of the overall decline in earnings. This decline in employer rank is related more to pay than changes in the composition or education levels of coworkers, as columns (3)–(5) show there are no significant changes in the fraction of natives or educated workers in the plant.

Taken in tandem, results suggest that getting stuck at lower paid employers, and to some extent lower paid service occupations, are important mechanisms behind why workers are worse off following the combined immigration and automation shocks.

Table 4: The Interaction Between Automation and Immigration, plant-level outcomes

	(1) Δ Plant Rank All Workers	(2) Δ Plant Rank Native Workers	(3) Δ Frac. Native	(4) Δ Frac. College Educ.	(5) Δ Frac. Comp. Educ.
Δ automation $\times$ treated	-1.293** (0.528)	-1.401** (0.585)	0.003 (0.003)	-0.003 (0.004)	-0.006 (0.005)
Education FE (1 digit)	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes
Individuals	78322	78322	78323	78323	78323
Average x	1.327	1.327	1.327	1.327	1.327

Notes: Table reports estimates from equation (4), with outcome variables measured at the plant-level. Columns (1) and (2) measure the change in the percentile rank of plants, for the median earnings of all workers and native workers respectively. Column (3) measures the change in the fraction of native workers. Column (4) measures the change in the fraction of college educated workers, while column (5) measures the change in the fraction of low educated workers with only compulsory schooling. Standard errors are clustered at the initial local labor market level. ***, **, and * correspond to significance at the 1%, 5%, and 10% levels respectively.

7 Conclusion

How do shocks which cause major changes in the labor market interact with each other? We show that two contemporaneous shocks, automation and immigration, interact with each other and these compound effects can exacerbate labor market challenges faced by the non-college educated. Workers who are exposed to both automation and immigration shocks see significantly higher earnings losses relative to workers exposed just to automation. The interaction between shocks is sufficiently large to affect the overall interpretation of the shocks when estimated in isolation. In the absence of one shock, the estimated effects of other shocks become smaller in magnitude among our sample of non-college educated workers. This interaction affects workers outside the manufacturing industries via spillovers, and earnings losses among workers exposed to both shocks are explained by shifts to lower paid employers and, to some extent, a shift from blue-collar occupations to lower paid services.

The objective of this paper is not to examine the potentially exhaustive set of interactions among all possible shocks which affect workers differently; instead, our results highlight the need for an integrated approach to comprehensively understand the key patterns which shape the labor market. While extensive literatures on automation and immigration present mixed results, we demonstrate that the interaction between different shocks can play an important role in understanding the divergence in findings. A unified approach is needed to inform any policy response to mitigate these negative labor market shocks, and an understanding of how different shocks interact with each other is imperative.

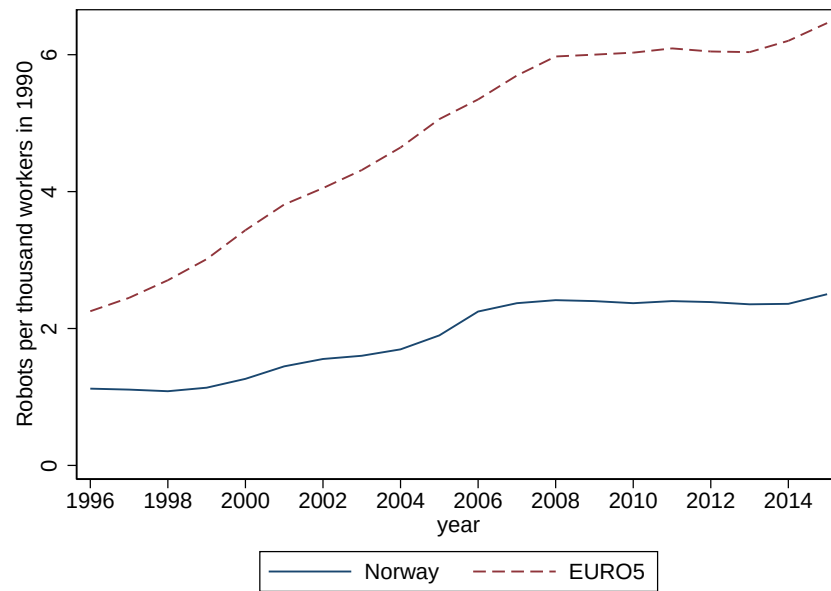
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A Data description: Norwegian Register Data and IFR Robotics

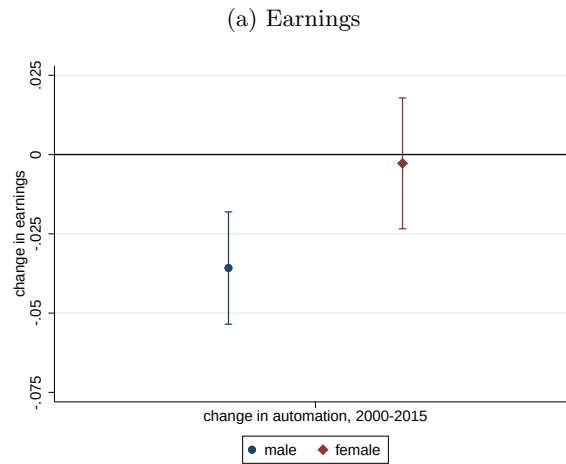
Figure A.1: The evolution of industrial robots over time



Notes: 19 industries we use in our measure of automation shock

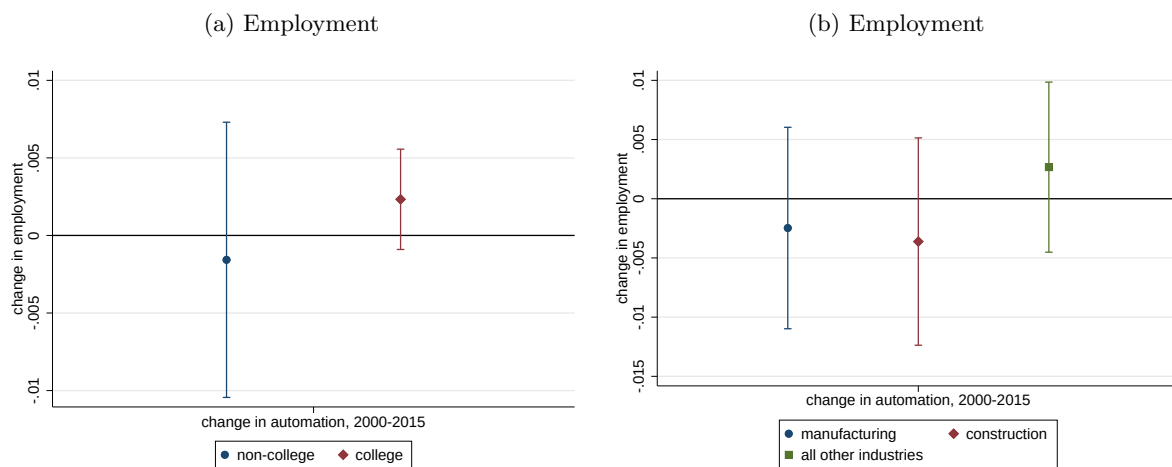
B The Effects of Automation on Labor Market Outcomes, by Gender, Industry, and Education

Figure B.2: The Impacts of Automation on Earnings by Gender, from 2000–2015



Notes: sample of men/women aged 18–49 employed initially in 2000. Figure presents estimates of equation (3), estimated separately for male/female workers.

Figure B.3: The Impacts of Automation on Employment, by Industry/Education, from 2000–2015



Notes: sample of men aged 18–49 employed initially in 2000.

C Comparing Workers in Manufacturing, Construction, and Other Industries

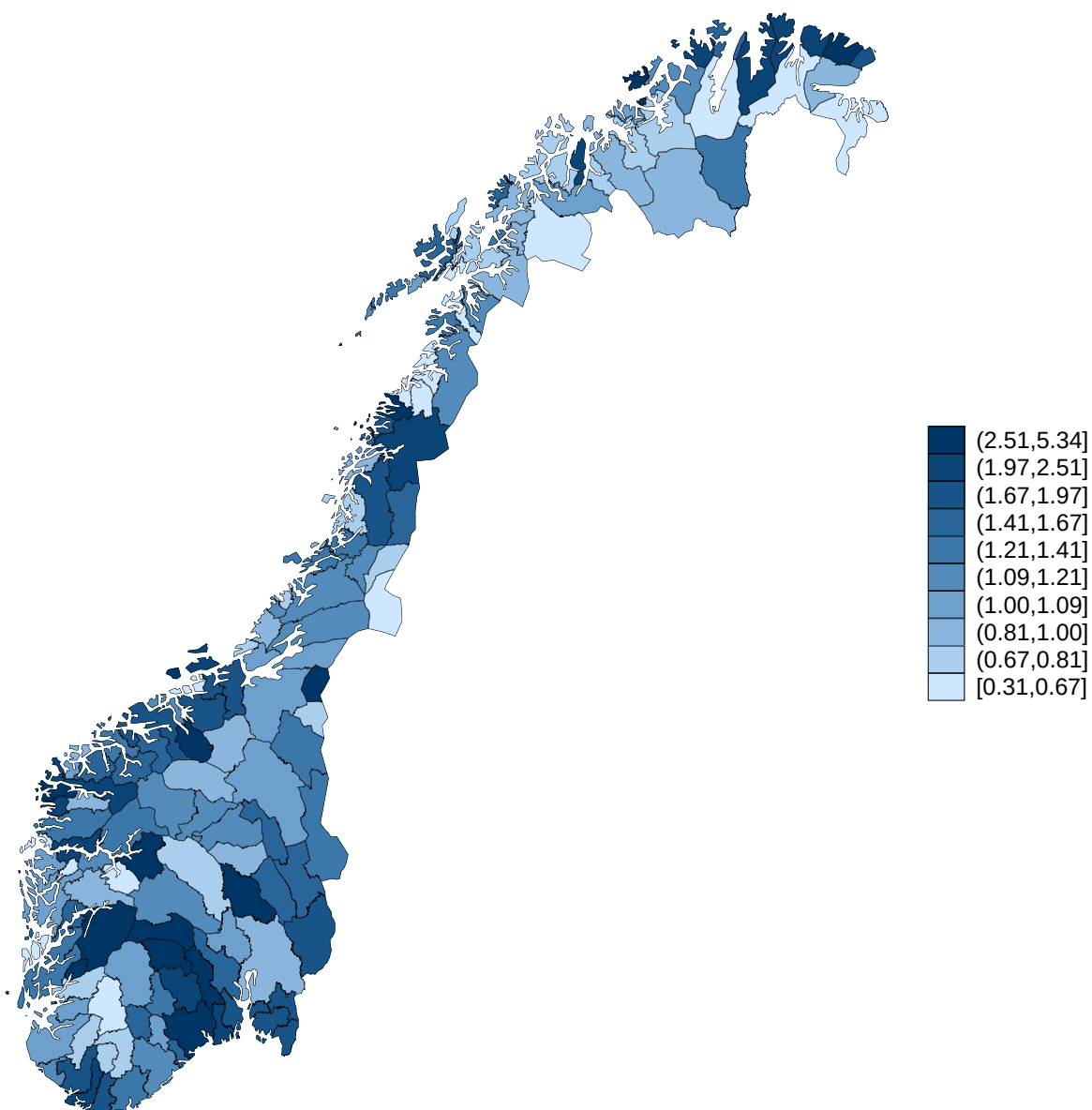
Table C.1: Comparing Construction, Manufacturing, and Other Industries

	(1) construction	(2) manufacturing	(3) all other industries
<i>Demographics:</i>			
male	0.911 (0.284)	0.740 (0.438)	0.466 (0.499)
norwegian	0.955 (0.207)	0.931 (0.254)	0.929 (0.257)
age	38.73 (11.92)	39.99 (11.76)	39.51 (12.15)
Lives Oslo or Bergen	0.257 (0.437)	0.223 (0.416)	0.342 (0.474)
<i>Labor Market:</i>			
log earnings	12.38 (0.613)	12.41 (0.641)	12.21 (0.782)
<i>Education:</i>			
HS grad	0.553 (0.497)	0.526 (0.499)	0.610 (0.488)
Compulsory education	0.285 (0.451)	0.288 (0.453)	0.208 (0.406)
University grad	0.0838 (0.277)	0.154 (0.361)	0.324 (0.468)
Individuals	144856	288809	1738161

Notes: sample of workers in one of three industries. All variable measured in year 2000.

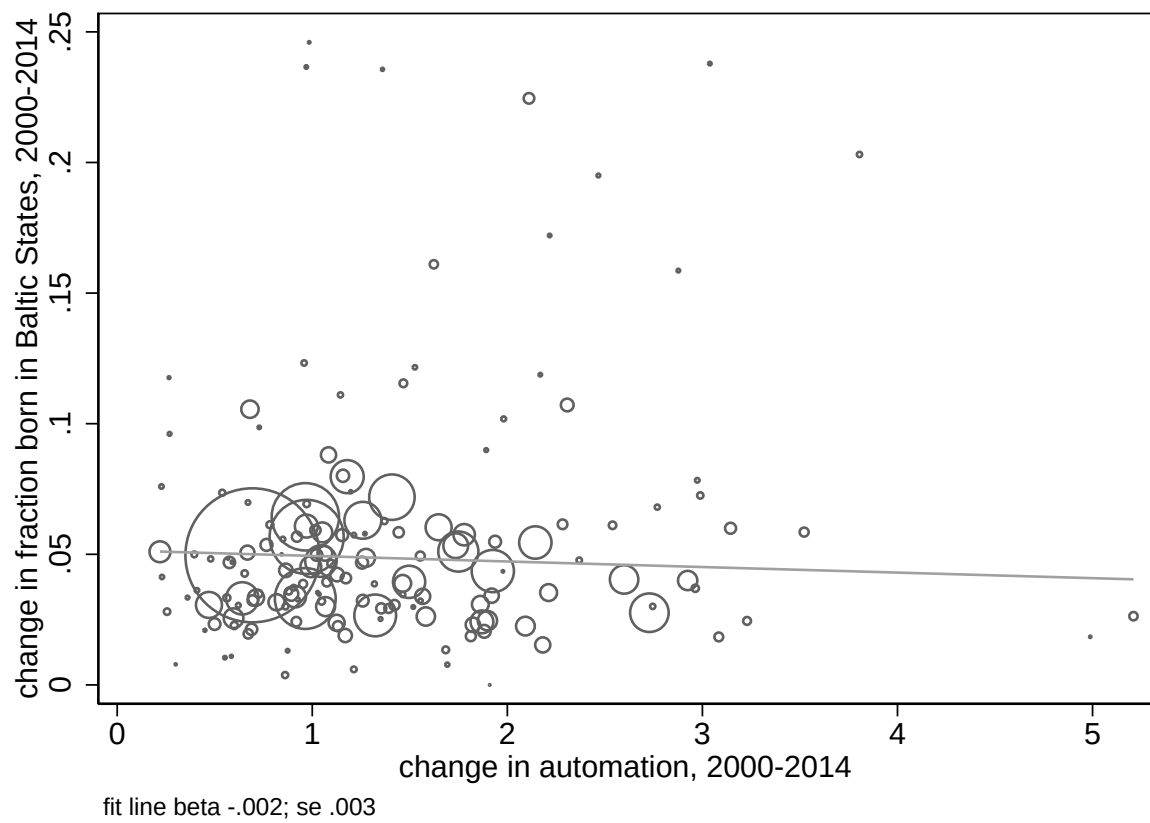
D The Spatial Distribution of The Automation Shock

Figure D.4: Describing the Local Labor Market Variation in Automation



Notes: Figure reports the exposure to robots (robots per 1000 workers) as defined as in Acemoglu and Restrepo (2020) from 1996–2015.

Figure D.5: The Relationship Between Automation and Immigration Shocks at the Area-Level



Notes: Figure plots the relationship between realized immigration (y-axis) and the change in automation (x-axis), both measured at the area-level. Each circle corresponds to a local labor market, where the size corresponds to the population. Both scatter plot and regression fit line (estimated at the area-level) are weighted by population.

E The Interacted Impacts of Automation and Immigration on Employment

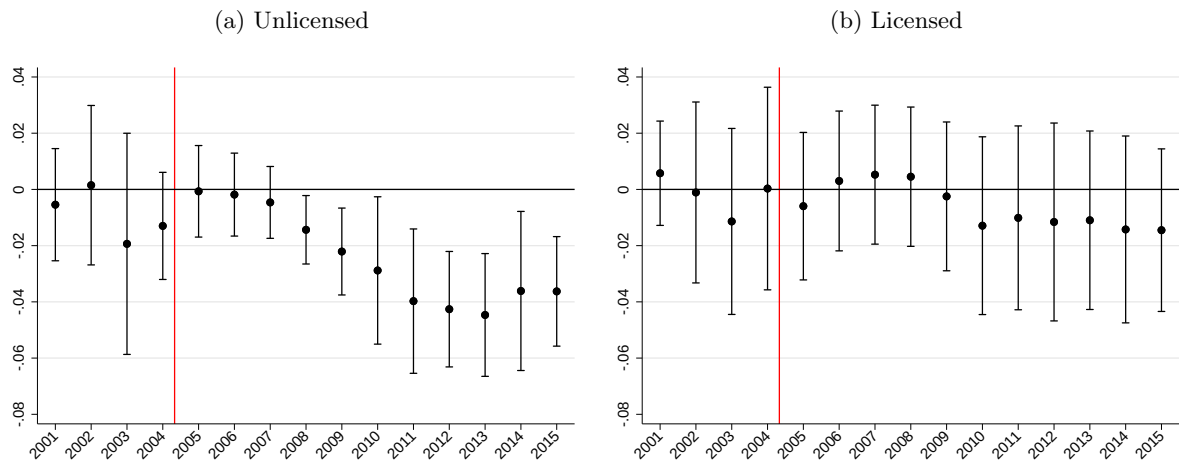
Table E.2: The Interaction Between Automation & Immigration, Employment

	Δ Employed		
	(1) Full Sample	(2) Licensed	(3) Unlicensed
Δ automation, 1996-2015	-0.000 (0.005)	-0.001 (0.004)	-0.001 (0.006)
Unlicensed 2000	-0.027*** (0.005)		
Δ automation $\times$ unlicensed 2000	-0.001 (0.004)		
Education FE (1 digit)	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes
Individuals	87512	43494	44018
Average x	1.328		

Notes: Columns (1) reports estimates from equation (4), for employment. Columns (2)–(3) report estimates from equation (3), separately for licensed and unlicensed workers, for employment. Standard errors are clustered at the initial local labor market level. Standard errors are clustered at the initial local labor market level. ***, **, and * correspond to significance at the 1%, 5%, and 10% levels respectively.

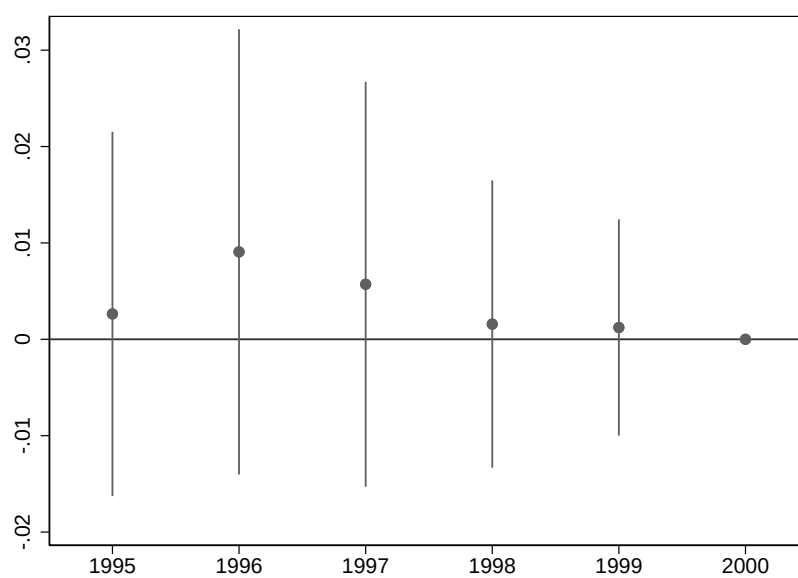
F Robustness

Figure F.6: The Impacts of Automation, Over Time (flexible shock) with Separate Difference-in-Differences



Notes: vertical line at 2004 represents when EU expansion occurs.

Figure F.7: The Interaction Between Automation and Immigration, Prior to Onset of Both Shocks



Notes: Figure plots the estimated interaction coefficients between automation and immigration shocks, prior to the onset of both shocks. Sample size becomes increasingly smaller going back further in time, as younger individuals employed in 2000 have not yet transitioned into the labor market 5 years earlier.